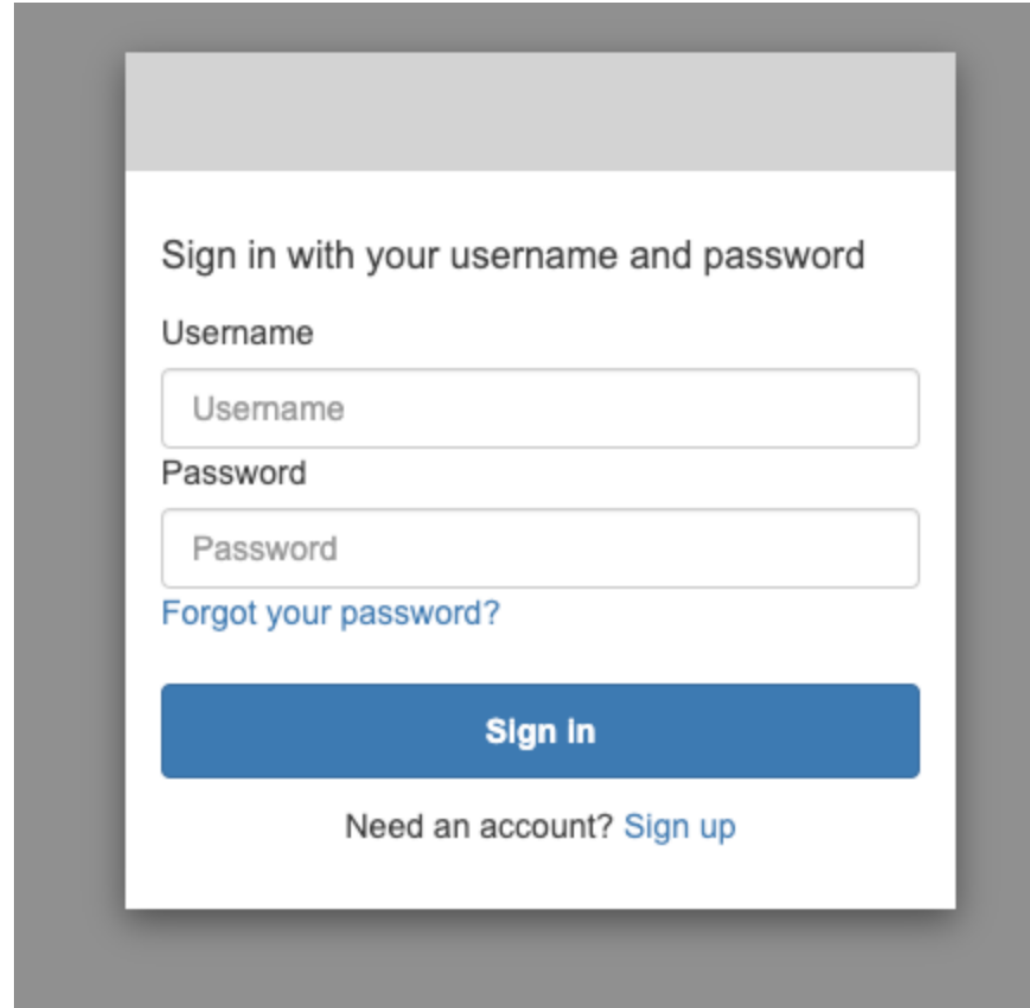


Step 1: Sign Up

Use this section if this is your first time connecting to the system.

1. Open your browser and navigate to TRE URL. <https://mbzuai-us.tre.pheno.ai/>
2. Click on the **Sign up** link.



Sign in with your username and password

Username

Password

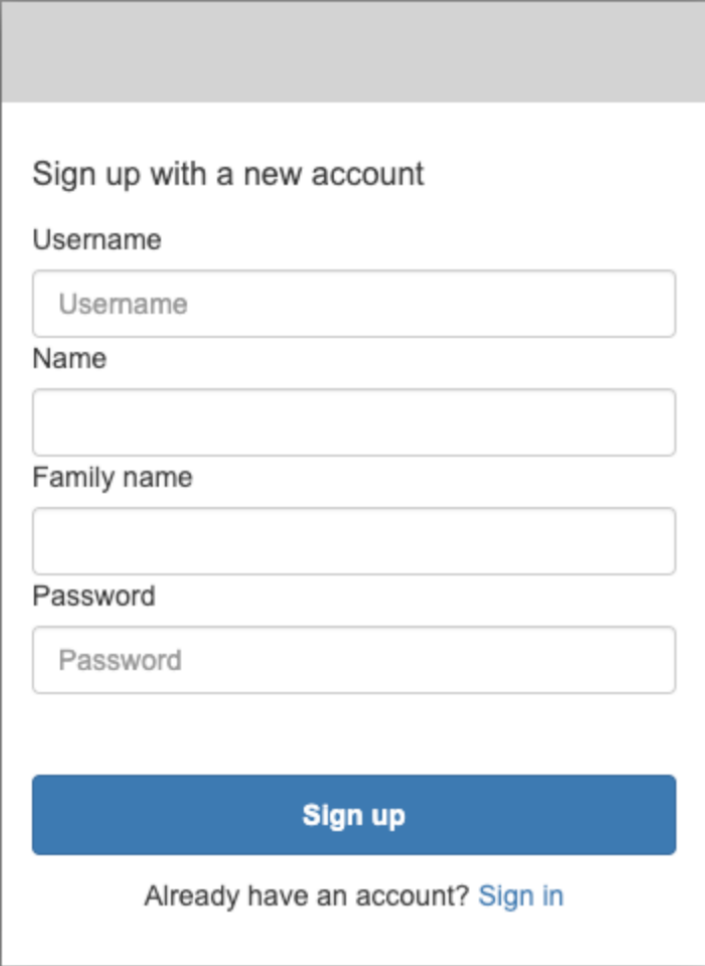
[Forgot your password?](#)

Sign In

Need an account? [Sign up](#)

Step 1: Sign Up

3. Fill out your **Username** (email address), **Name**, **Family name** and pick a **Password**.



Sign up with a new account

Username

Name

Family name

Password

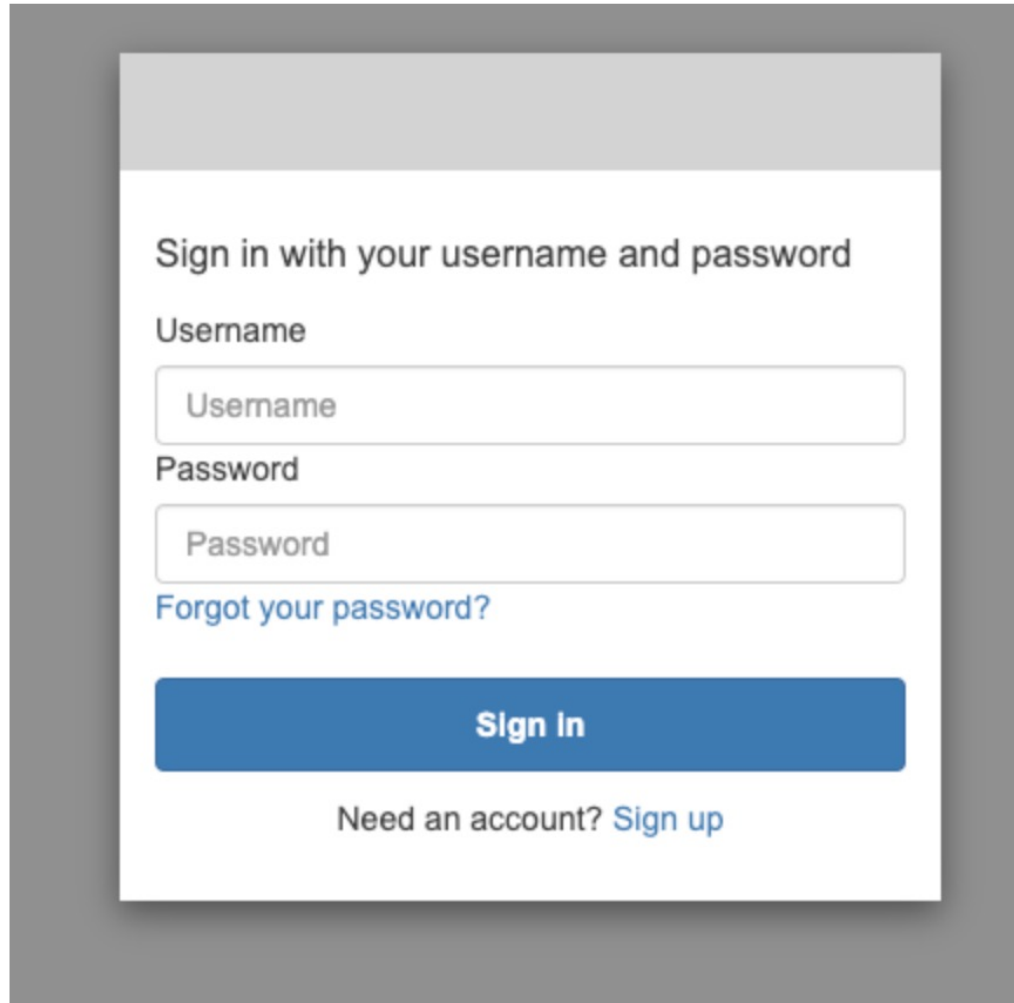
Sign up

Already have an account? [Sign in](#)

4. Click the **Sign up** button.

Step 2: Login or Sign up

1. Open your browser and navigate to TRE URL.
2. Type in your **Username** (the email you used to sign up) and your **Password**.
3. Click the **Sign In** button.

A login form is displayed on a white background with a light gray header bar at the top. The form contains the text "Sign in with your username and password" in a dark gray font. Below this text are two input fields: the first is labeled "Username" and the second is labeled "Password". Both fields have a light gray border and placeholder text. Below the password field is a blue link that says "Forgot your password?". At the bottom of the form is a blue button with the text "Sign In" in white. Below the button is the text "Need an account?" followed by a blue link that says "Sign up".

Sign in with your username and password

Username

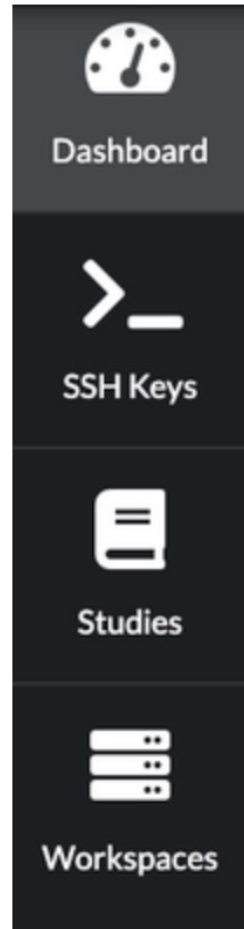
Password

[Forgot your password?](#)

Sign In

Need an account? [Sign up](#)

Interface



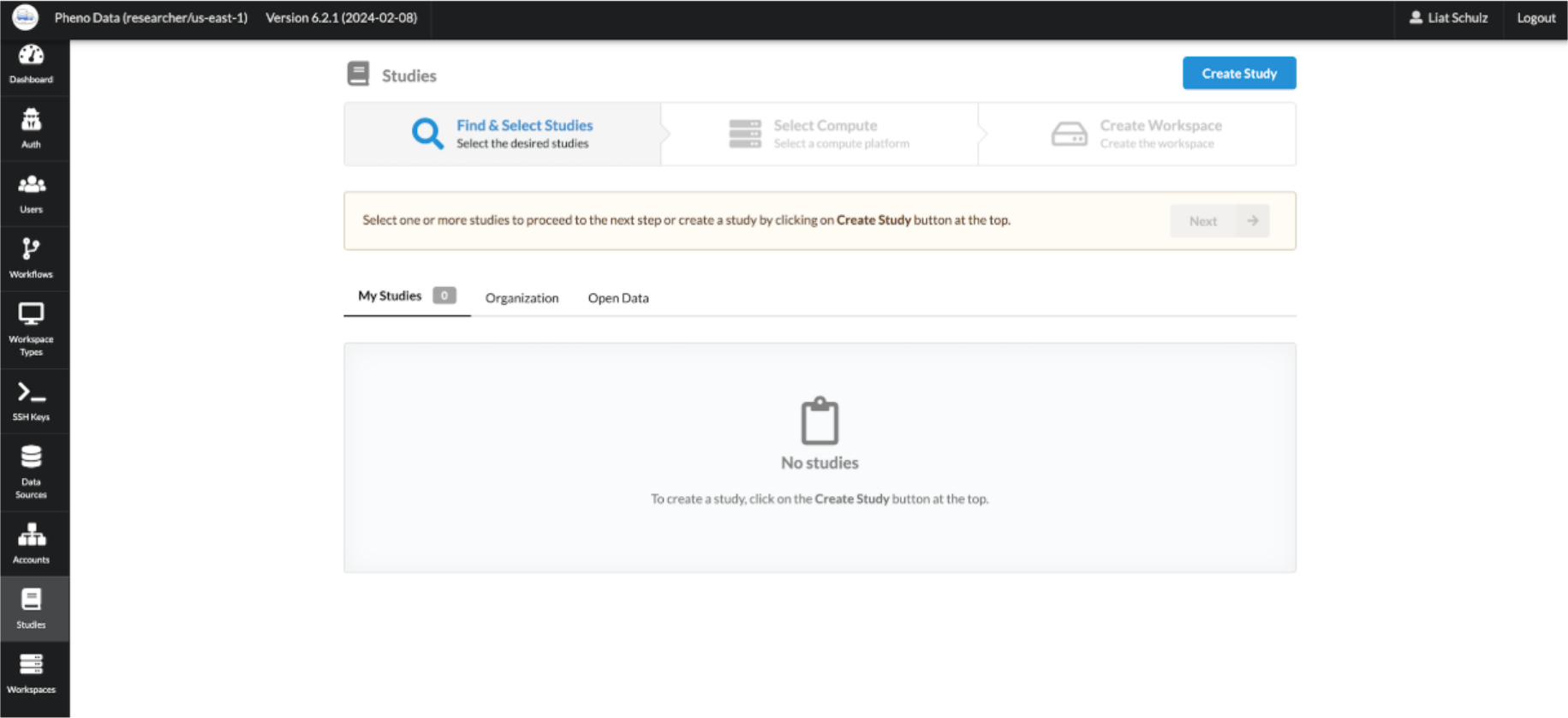
Dashboard - Displays AWS charges over the past 30 days

SSH Keys - SSH Keys for EC2 authorization

Workspaces - The combination of a study with a compute

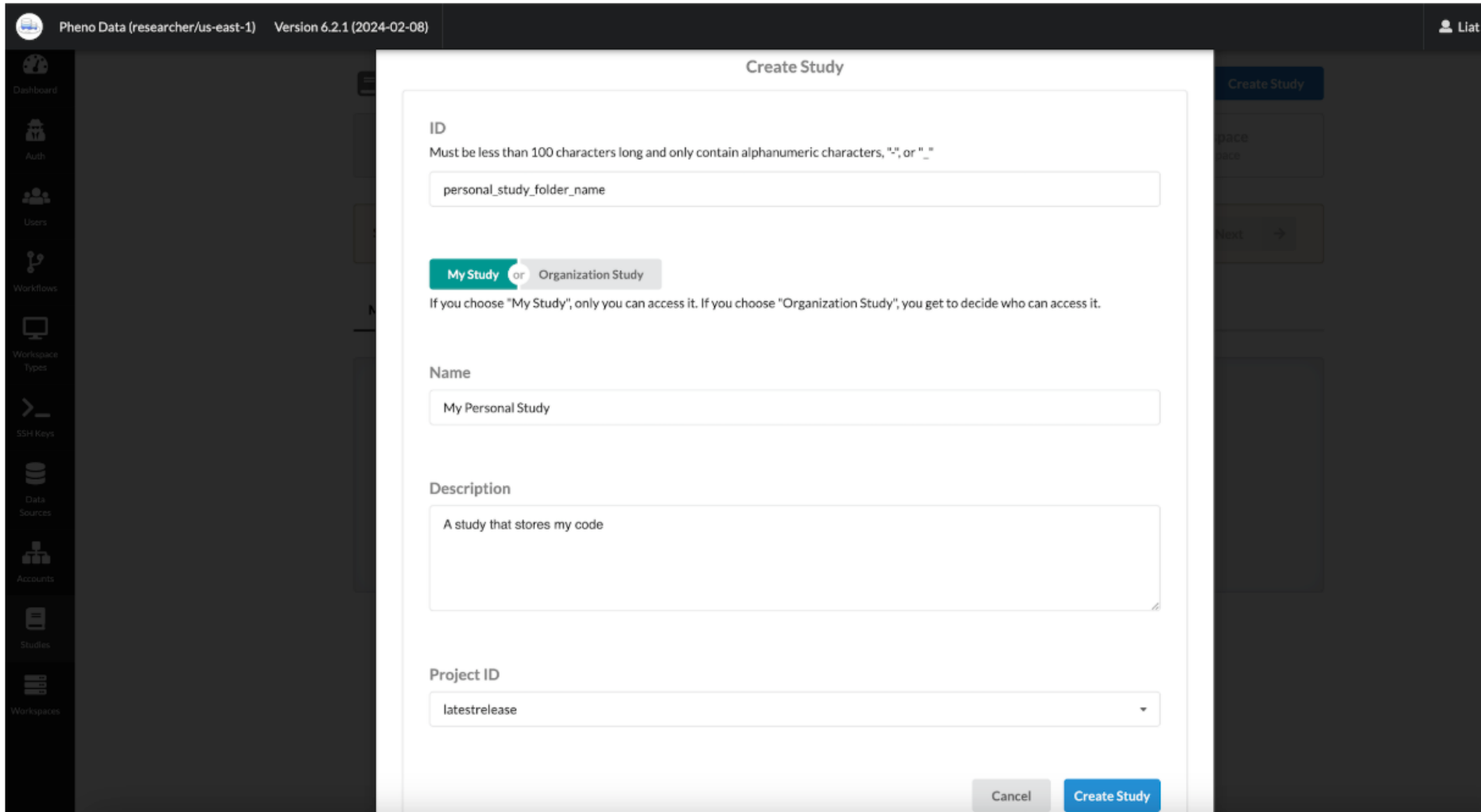
Step 3: Begin your study

1. Click on Create Study



Step 3: Begin your study

2. Fill up all the fields and press **Create Study** myStudy for personal and Organization study for Org Studies



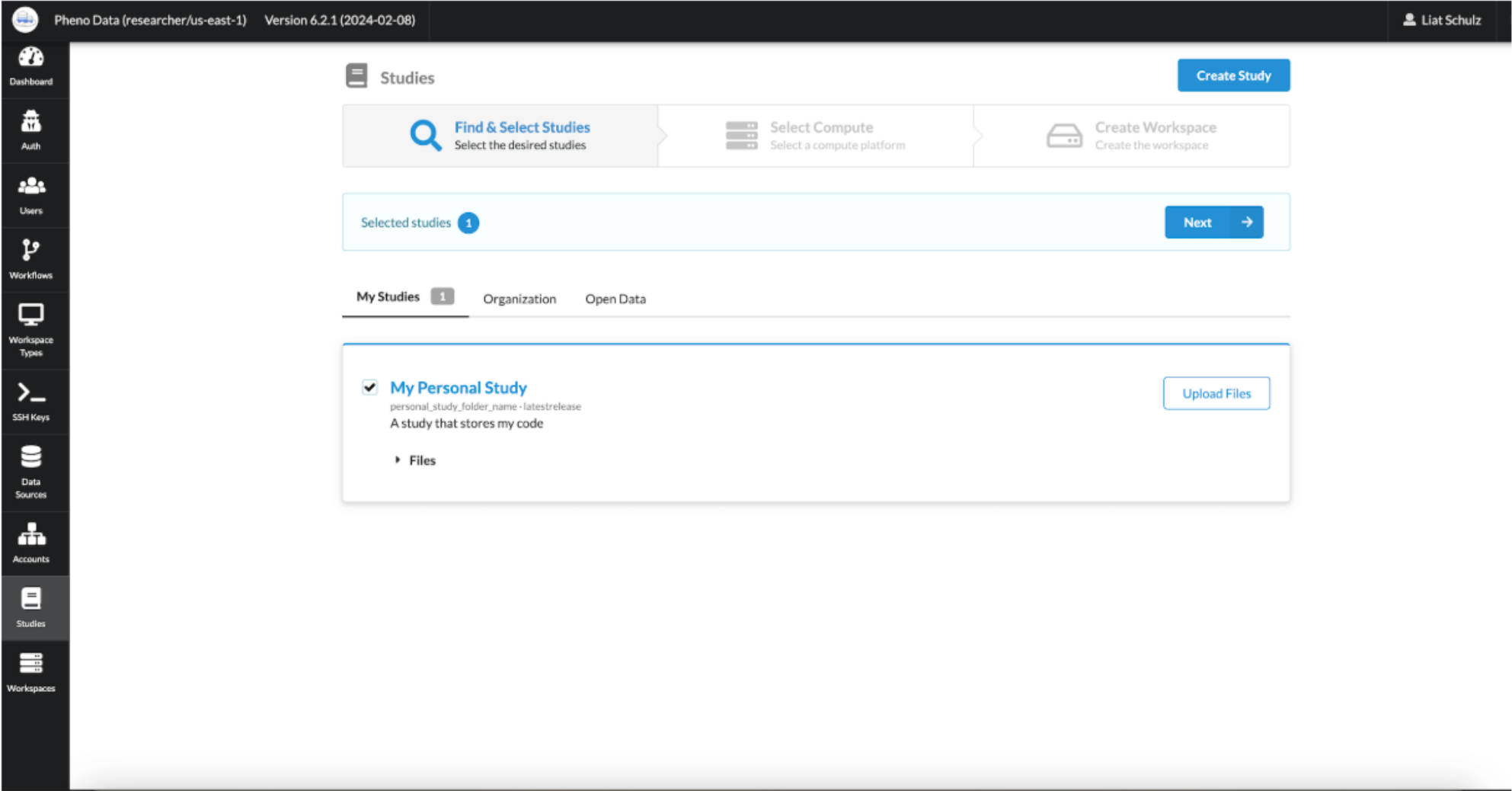
The screenshot shows the 'Create Study' dialog box in the Pheno Data application. The dialog is titled 'Create Study' and contains the following fields and options:

- ID**: A text input field with the placeholder 'personal_study_folder_name'. Above the field, it states: 'Must be less than 100 characters long and only contain alphanumeric characters, "-", or "_"'. To the right of the dialog, a 'Create Study' button is visible.
- Study Type**: Two radio buttons labeled 'My Study' (selected) and 'Organization Study'.
- Instructions**: A line of text below the radio buttons: 'If you choose "My Study", only you can access it. If you choose "Organization Study", you get to decide who can access it.'
- Name**: A text input field with the placeholder 'My Personal Study'.
- Description**: A text area with the placeholder 'A study that stores my code'.
- Project ID**: A dropdown menu with the selected value 'latestrelease'.
- Buttons**: At the bottom right, there are two buttons: 'Cancel' and 'Create Study'.

The background interface shows a sidebar with navigation icons for Dashboard, Auth, Users, Workflows, Workspace Types, SSH Keys, Data Sources, Accounts, Studies, and Workspaces. The top header displays 'Pheno Data (researcher/us-east-1) Version 6.2.1 (2024-02-08)' and a user profile icon labeled 'Liat S'.

Step 3: Begin your study

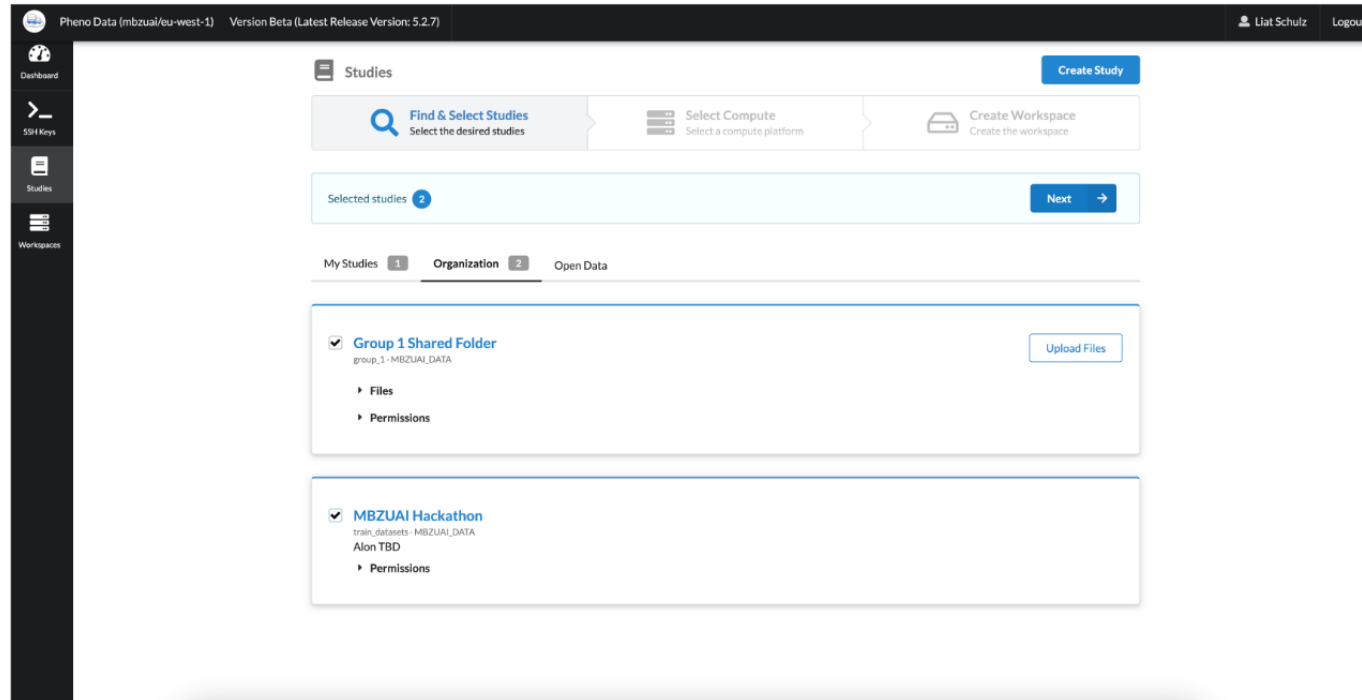
3. Study Created and ready to use



Step 4: Launching a new Research Environment on the study (study links the disk with the research data, and Research Environment gives you the environment to work on the data)

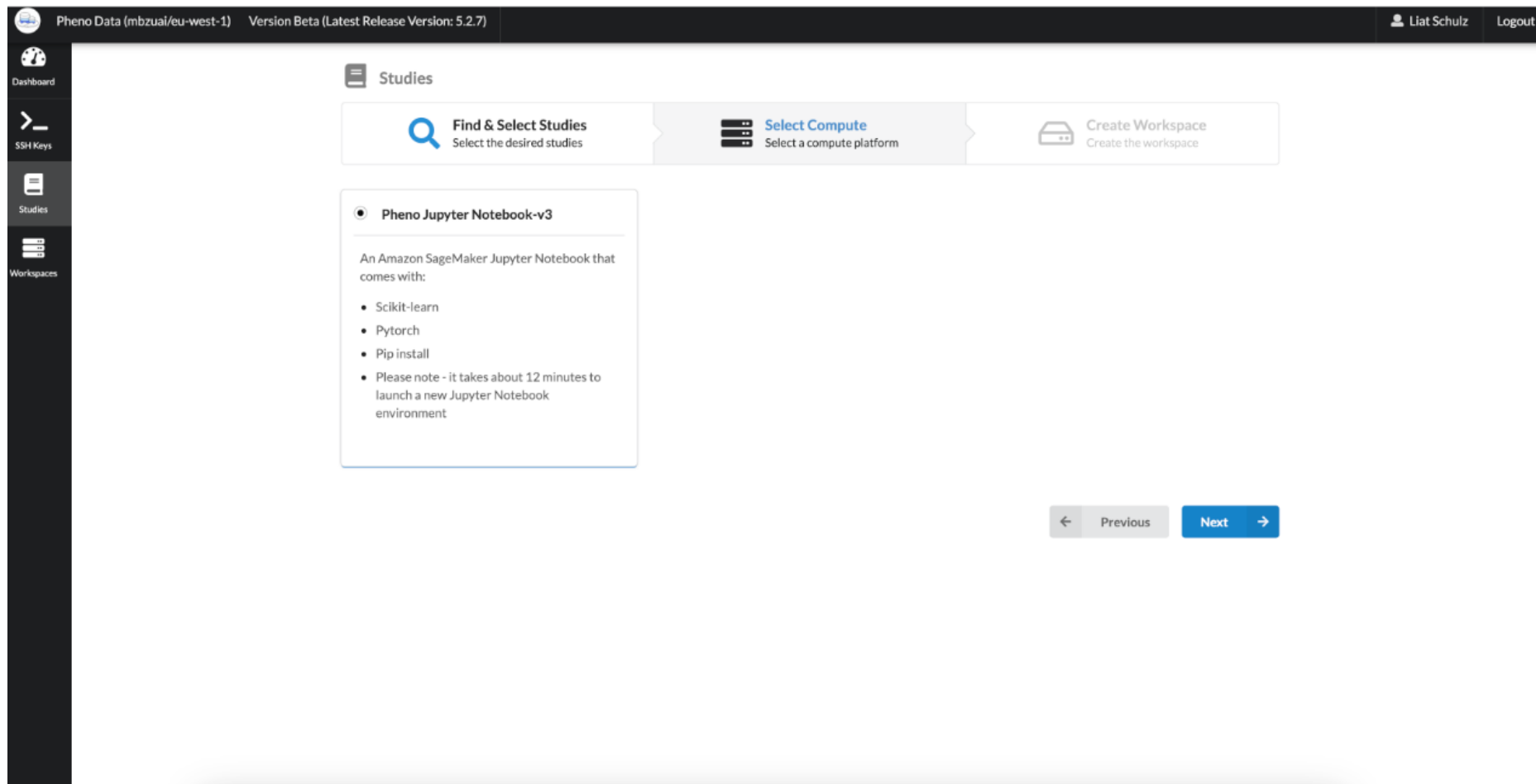
A user can select a Study or multiple Studies and launch a Workspace to access and analyze data. To launch a Research Environment, follow these steps:

1. In the portal navigate to the **Studies** page using the menu on the left.
2. Select the Studies to be attached to the new Workspace.
3. Once you have selected all the Studies you want, click the **Next** button.



Step 4: Launching a new Research Environment on the study

4. Choose the type of Workspace you want and click the **Next** button.



The screenshot displays the Pheno Data web interface. At the top, a dark header bar contains the text 'Pheno Data (mbzual/eu-west-1) Version Beta (Latest Release Version: 5.2.7)' on the left and a user profile 'Liat Schulz' with a 'Logout' button on the right. A vertical sidebar on the left lists navigation options: 'Dashboard', 'SSH Keys', 'Studies' (which is highlighted), and 'Workspaces'. The main content area is titled 'Studies' and features a three-step process bar: 'Find & Select Studies' (active, with a magnifying glass icon), 'Select Compute' (with a server icon), and 'Create Workspace' (with a folder icon). Below this bar, a card titled 'Pheno Jupyter Notebook-v3' is shown, containing the text 'An Amazon SageMaker Jupyter Notebook that comes with:' followed by a bulleted list: 'Scikit-learn', 'Pytorch', 'Pip install', and 'Please note - it takes about 12 minutes to launch a new Jupyter Notebook environment'. At the bottom right, there are two buttons: 'Previous' and 'Next' (highlighted in blue).

Step 4: Launching a new Research Environment on the study

5. Fill in the **Workspace Name** field.
6. Select your project in the **Project ID** drop down field.
7. Select the **Configuration** type according to your needs.
8. Type a description for the Workspace in the **Description** field.
9. Click the **Create Research Workspace** button.

The screenshot displays the 'Pheno Data' web interface. The top navigation bar includes the user profile 'Liat Schulz' and a 'Logout' button. A sidebar on the left contains icons for Dashboard, Auth, Users, Workflows, Workspace Types, SSH Keys, Data Sources, Accounts, Studies, and Workspaces. The main content area is titled 'Studies' and features three primary action buttons: 'Find & Select Studies' (with a magnifying glass icon), 'Select Compute' (with a server rack icon), and 'Create Workspace' (with a laptop icon). Below these buttons, the 'Pheno Jupyter Notebook-v3' configuration form is shown. It includes a 'Name' field with a placeholder 'group1-liat-schulz-workspace1', a 'Project ID' dropdown menu set to 'MBZUAI_DATA', and a 'Configuration' section with three radio button options: 'Pheno-JupyterNotebookGeneralUsage' (selected), 'Pheno-JupyterNotebook1GPU', and 'Pheno-JupyterNotebook4GPU'. Each configuration option provides details on its estimated cost and instance specifications.

Configuration	Instance Type	Memory	Hourly Cost	Additional Info
☒ Pheno-JupyterNotebookGeneralUsage Jupyter Notebook General Usage	ml.t3.large	8 GiB	\$0.109	The Costs do not include the storage charges
☐ Pheno-JupyterNotebook1GPU Pheno Jupyter Notebook 1 GPU	ml.g5.2xlarge	32 GiB	\$1.69	The Costs do not include the storage charges
☐ Pheno-JupyterNotebook4GPU Pheno Jupyter Notebook 4 GPU	ml.g5.12xlarge	192 GiB	\$7.92	The Costs do not include the storage charges

Step 4: Launching a new Research Environment on the study

This will deploy the new Workspace and attach the Studies that were selected.

You will automatically be redirected to the Workspaces tab on the portal.

Wait for the workspace to be **Available**.

Pheno Data (mbzuai/eu-west-1) Version Beta (Latest Release Version: 5.2.7)

Liat Schulz Logout

Dashboard

SSH Keys

Studies

Workspaces

Configuration

☒ **Pheno-JupyterNotebookGeneralUsage**

Pheno Jupyter Notebook General Usage

Estimated Cost

Instance type ml.t3.large
VPCU 2
Memory 8 GiB Hourly Cost \$0.109

- The Costs do not include the storage charges

Instance Type ml.t3.large

Hardcoded CIDR

☐ **Pheno-JupyterNotebook1GPU**

Pheno Jupyter Notebook 1 GPU

Estimated Cost

Instance type ml.g5.2xlarge VPCU 8
Memory 32 GiB Hourly Cost \$1.69

- The Costs do not include the storage charges

Instance Type ml.g5.2xlarge

Hardcoded CIDR

☐ **Pheno-JupyterNotebook4GPU**

Pheno Jupyter Notebook 4 GPU

Estimated Cost

Instance type ml.g5.12xlarge
GPU 4 VPCU 48
Memory 192 GiB Hourly Cost \$7.92

- The Costs do not include the storage charges

Instance Type ml.g5.12xlarge

Hardcoded CIDR

Description

hackathon liat group 1

← Previous

Create Research Workspace

Step 5: Connect to the Research Environment

1. In the portal navigate to the **Workspaces** page using the menu on the left.
2. Click on the **Connections** button, below the **Workspace** name.

Pheno Data (mbzuai/eu-west-1) Version Beta (Latest Release Version: 5.2.7) Liat Schulz Logout

Research Workspaces 91 Create Research Workspace

All Available Pending Stopped Errored Terminated

AVAILABLE

group1-liat-schulz-workspace1
Created 4 hours ago by Liat Schulz c02327b6-13c8-42b8-9144-064442c62fbd

Connections View Detail Egress Store Stop Terminate

bioconda test

Owner	Liat Schulz
Studies	3
Project	MBZUAI_DATA
Workspace Type	Pheno Jupyter Notebook-v3
Configuration Name	Pheno-JupyterNotebookGeneralUsage
Instance Type	ml.t3.large

\$0.00
YESTERDAY'S COST

STOPPED

3. Click on the **Generate URL** button on the right, under HTTP Connections.
4. Click **Connect**.

Step 5: Connect to the Research Environment

3. Click on the **Generate URL** button on the right, under HTTP Connections.
4. Click **Connect**.

The screenshot displays the 'Research Workspaces' interface. The top navigation bar includes 'Pheno Data (mbzual/eu-west-1)', 'Version Beta (Latest Release Version: 5.2.7)', and a user profile 'Liat Schulz' with a 'Logout' button. A left sidebar contains icons for Dashboard, Auth, Users, Workflows, Workspace Types, SSH Keys, Data Sources, Accounts, Studies, and Workspaces. The main content area shows a list of workspaces, with the first one, 'group1-liat-schulz-workspace1', highlighted in green and labeled 'AVAILABLE'. This workspace was created 23 minutes ago by Liat Schulz. Below the workspace name are buttons for 'Connections', 'View Detail', 'Egress Store', 'Stop', and 'Terminate'. A section titled 'Connection instructions for your AppStream workspace:' provides a list of steps: note the 12-minute launch time, wait for the green 'AVAILABLE' status, click 'Generate URL', copy the URL, paste it in a new AppStream Chrome tab, and allow popups. Below this, the 'HTTP Connections' section shows a 'SageMaker Notebook' with a 'Generate URL' button. A link to copy the destination URL is provided, followed by a 'Connect' button. At the bottom, a table shows the workspace owner as 'Liat Schulz'.

Pheno Data (mbzual/eu-west-1) Version Beta (Latest Release Version: 5.2.7) Liat Schulz Logout

Research Workspaces 91 Create Research Workspace

All Available Pending Stopped Errored Terminated

AVAILABLE

group1-liat-schulz-workspace1
Created 23 minutes ago by Liat Schulz c02327b6-13c8-42b8-9144-064442c62fbd

Connections View Detail Egress Store Stop Terminate

Connection instructions for your AppStream workspace:

- Please note it takes around 12 minutes to launch a new environment
- Wait till the environment marked with green as available
- Click the "Generate URL" button to create the destination URL
- Copy the URL and hit "Connect".
- Paste the URL in the new AppStream Chrome tab
- In your browser, please allow popups for this domain so we can open the AppStream page in a new tab for you

HTTP Connections

SageMaker Notebook Generate URL

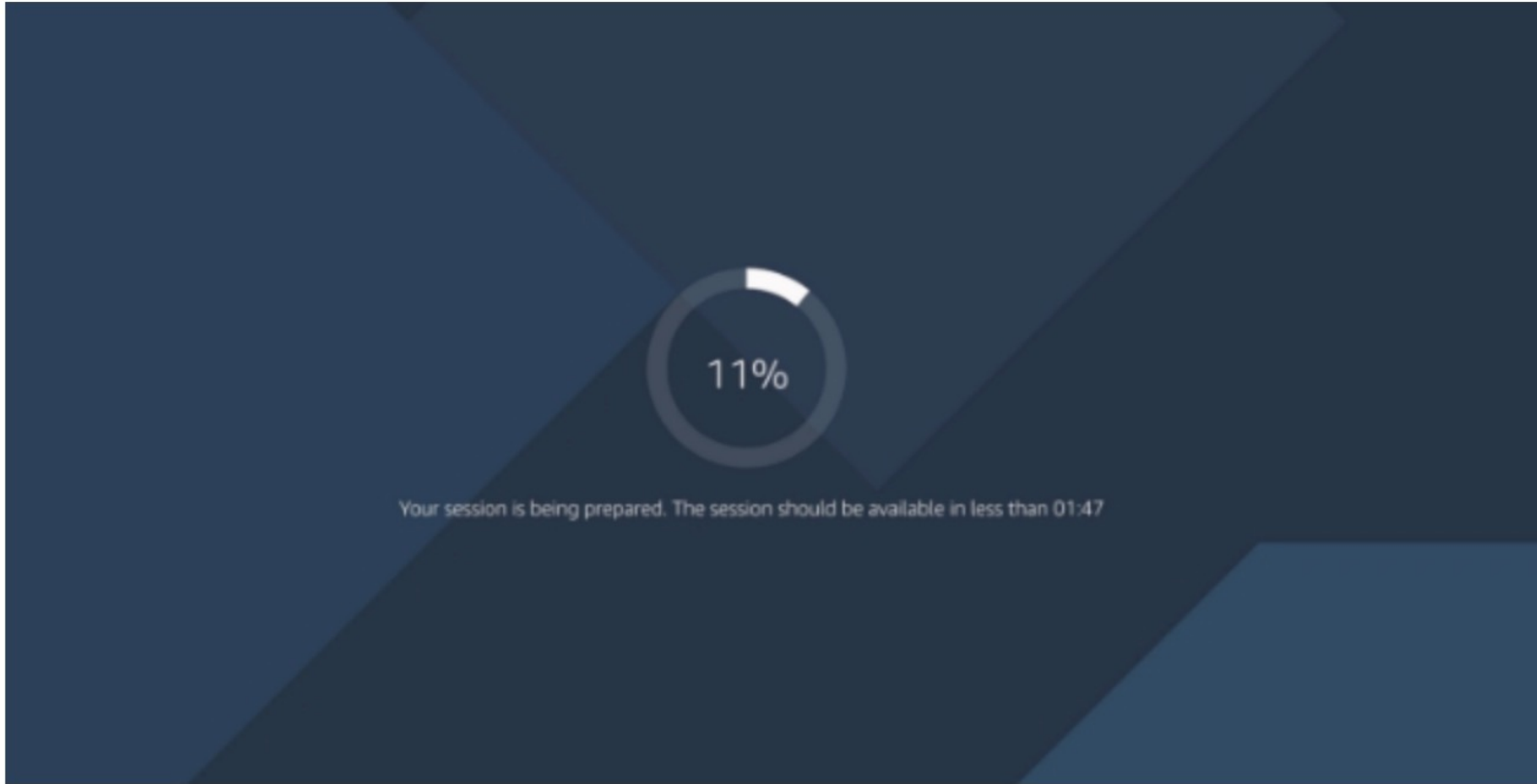
Click here to copy the destination URL: Connect

bioconda test

Owner	Liat Schulz
-------	-------------

Step 5: Connect to the Research Environment

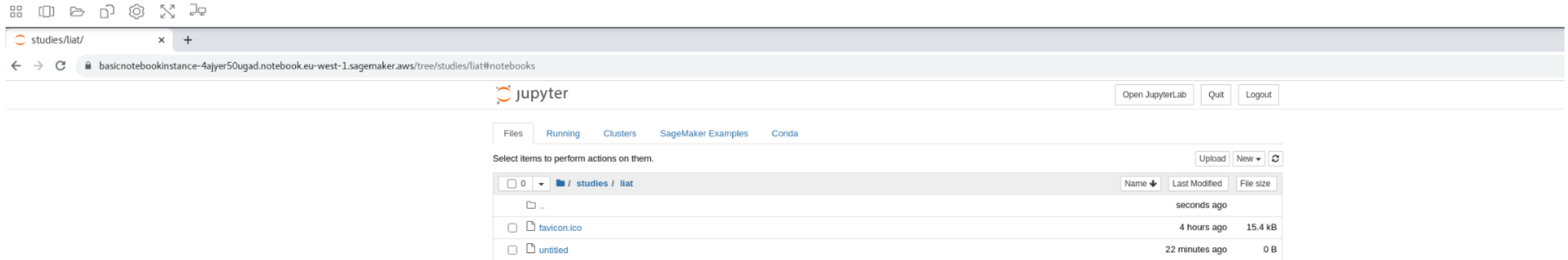
5. Wait for the system to load.



It will now open the browser with the data and the tools to start working. It will open in Firefox automatically if the workspace created in a Notebook, or Remmina Client(Remote) in case it created in EC2. So you can do the following:

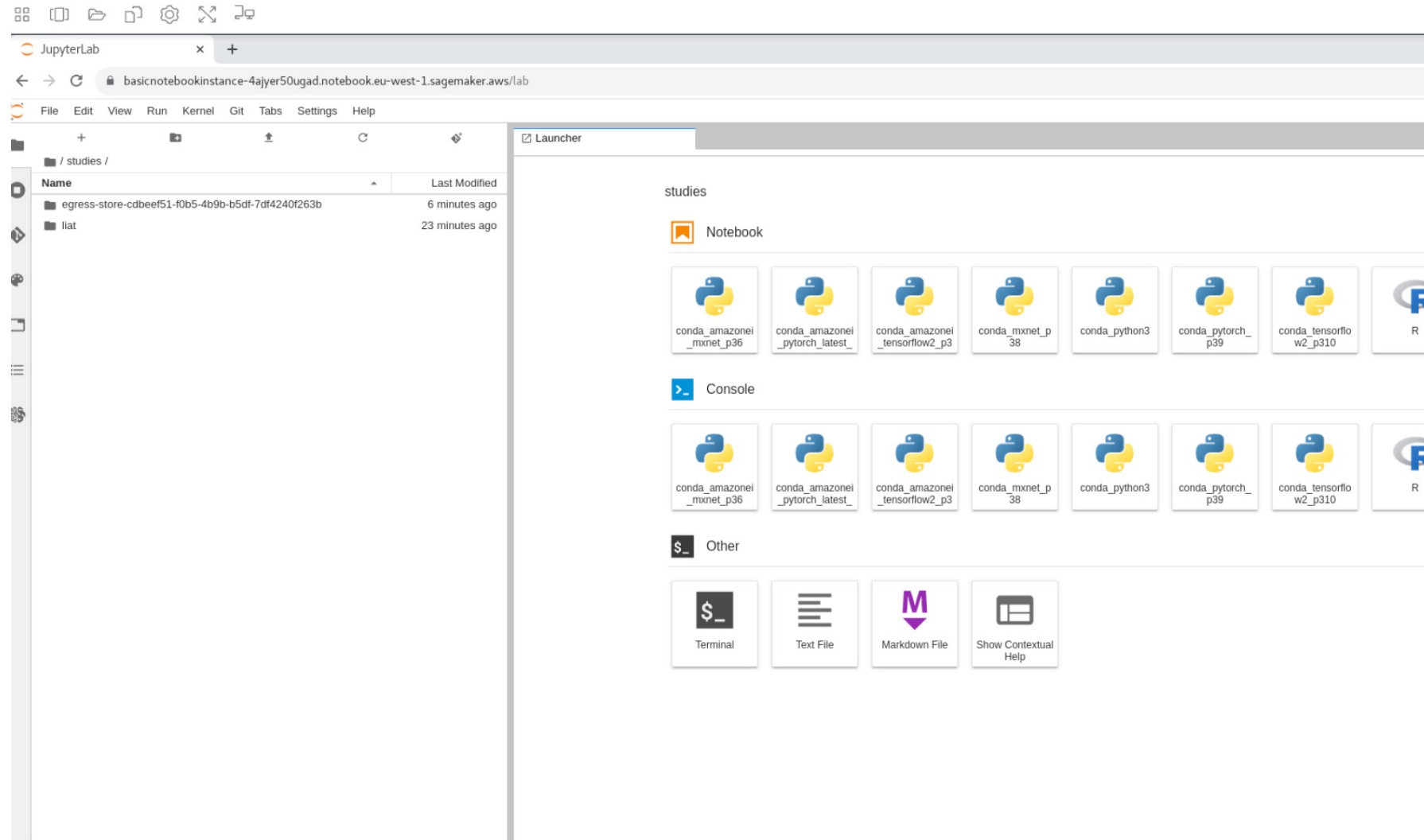
Step 6: Installing Python Packages to the Research Environment

1. Connect to the Research Environment and open your Jupyter Notebook (See: Connect to an Existing Research Environment)
2. Click on Open JupyterLab on the top right.



Step 6: Installing Python Packages to the Research Environment

3. Open a new Jupyter notebook.



4. Inside the one of the notebook cells, type `!pip3 install <package name>`.

Step 7: Attach Files to a Research Environment

1. Navigate to the **Studies** page using the menu on the left.
2. Find the Study you want to attach files to and click **Upload Files** on the right.
3. You can upload files either by dragging and dropping, or by clicking the **Upload Files** or **Upload Folder** button.

The screenshot displays the Pheno Data web application interface. At the top, a dark header bar contains the text "Pheno Data (mbzuai/eu-west-1)" and "Version Beta (Latest Release Version: 5.2.7)" on the left, and a user profile icon with the name "Liat Schulz" and a "Logout" link on the right. A vertical sidebar on the left lists navigation options: Dashboard, Auth, Users, Workflows, Workspace Types, SSH Keys, Data Sources, Accounts, Studies (highlighted), and Workspaces. The main content area is titled "Studies" and features a "Create Study" button in the top right. Below the title, there are three steps: "Find & Select Studies" (with a magnifying glass icon and the text "Select the desired studies"), "Select Compute" (with a server icon and the text "Select a compute platform"), and "Create Workspace" (with a folder icon and the text "Create the workspace"). A yellow instruction box states: "Select one or more studies to proceed to the next step or create a study by clicking on Create Study button at the top." with a "Next" button and a right arrow. Below this, there are three tabs: "My Studies" (with a count of 1), "Organization" (with a count of 3), and "Open Data". The "Organization" tab is active, showing a list of studies. The first study is "BioConda" (bioconda · MBZUAI_DATA) with a checkbox and a "Permissions" link. The second study is "Group 1 Shared Folder" (group_1 · MBZUAI_DATA) with a checkbox, a "Permissions" link, and an "Upload Files" button. The third study is "MBZUAI Hackathon" (train_datasets · MBZUAI_DATA) with a checkbox, a "Permissions" link, and the text "Human Phenotype Project datasets".

Step 8: Tools Installation

Docker pull

docker registry: <https://gallery.ecr.aws/>

Use the following code `docker pull \<accountID>.dkr.ecr.\<aws-region>.amazonaws.com/ecr-public/\<Change to your library>`

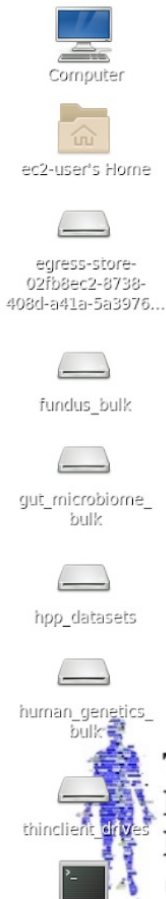
[illegible]

Step 8: Tools Installation

Docker run

Use the image you have under docker image ls

e.g `docker run -it --rm <accountID>.dkr.ecr.<aws-region>.amazonaws.com/ecr-public/docker/library/alpine:edge`



```
ec2-user@ip-10-0-93-81:~$  
File Edit View Search Terminal Help  
/efs is already mounted, please run 'mount' command to verify  
In order to pull from docker from ECR please execute /home/ec2-user/scripts/docker_pull.sh  
In need of pulling docker use the following syntax: docker pull 104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/<Change to your library>  
e.g 000000000.dkr.ecr.us-east-1.amazonaws.com/ecr-public/amazonlinux/amazonlinux  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$ docker pull 104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/docker/library/alpine:edge  
edge: Pulling from ecr-public/docker/library/alpine  
Digest: sha256:e31c3b1cd47718260e1b6163af0a05b3c428dc01fa410baf72ca8b8076e22e72  
Status: Image is up to date for 104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/docker/library/alpine:edge  
104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/docker/library/alpine:edge  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$ docker image ls  
REPOSITORY                                TAG      IMAGE ID      CREATED      SIZE  
104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/docker/library/alpine    edge     49b3cb3043cd  4 weeks ago  7.74MB  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$  
[ec2-user@ip-10-0-93-81 ~]$ docker run -it --rm 104399116944.dkr.ecr.us-east-1.amazonaws.com/ecr-public/docker/library/alpine:edge  
/#  
/# ls  
bin  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var  
/# exit  
[ec2-user@ip-10-0-93-81 ~]$
```